

Notes: Yang (RES, 2020)
Optimality of Debt under Flexible Information Acquisition

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1 Big picture

- **Question.** Why is debt the optimal security when a seller wants liquidity but the buyer can flexibly acquire costly information about the underlying asset before accepting the offer?
- **Core mechanism.** Security shape determines what the buyer finds useful to learn. A payoff that is highly sensitive to fundamentals gives the buyer incentives to acquire information and cherry-pick states.
- **Main result.** In the baseline fixed-aggregate-cash-flow model, debt is uniquely optimal because it is the least information-sensitive feasible security.
- **Economic takeaway.** Debt is optimal not because it maximizes risk sharing, but because it minimizes information sensitivity and thereby mitigates adverse selection by sophisticated buyers.

2 Model primitives and measurement

2.1 Seller, buyer, and timing

A risk-neutral seller owns an asset in place that generates cash flow θ . The seller is impatient and issues a security s backed by the asset at a proposed price q . The buyer observes the offer, chooses a costly information structure, and decides whether to accept. The baseline model has fixed aggregate cash flow: the cash flow exists regardless of whether trade occurs.

Interpretation. This assumption makes information acquisition redistributive rather than surplus creating.

2.2 Feasible securities and debt

A feasible security satisfies

$$0 \leq s(\theta) \leq \theta.$$

A debt security with face value D is

$$s^D(\theta; D) = \min\{\theta, D\}.$$

Interpretation. The 45-degree region is imposed by limited liability; the flat tail is chosen to reduce information sensitivity.

2.3 Buyer payoff and information acquisition

The buyer's gain from accepting is

$$\Delta u(\theta) = \delta_b s(\theta) - q.$$

Let $m(\theta)$ denote the probability that the buyer accepts in state θ , and let $p_1 = E[m(\theta)]$. Under the binary-action representation, the buyer's interior optimality condition is

$$\Delta u(\theta) = \mu [g'(m(\theta)) - g'(p_1)].$$

Interpretation. The buyer raises acceptance probability in high-gain states and lowers it in low-gain states until marginal information cost equals marginal value.

2.4 Seller objective

The seller chooses s and q to maximize

$$W(s, q) = E[m_{s,q}(\theta)(q - \delta_s s(\theta))].$$

Interpretation. The seller internalizes that changing the security shape changes the buyer's information acquisition and acceptance rule.

3 Models and theoretical specifications

3.1 Positive trade and non-pass-through: Proposition 2

At the optimum, trade occurs with positive probability and the optimal security is not a pass-through claim. A pass-through claim makes the security maximally sensitive to θ and gives the buyer strong incentives to acquire information.

3.2 Debt without information acquisition: Proposition 3

If the optimal security at price q induces no information acquisition, then it is debt:

$$s_q(\theta) = \min\{\theta, D^*(q)\}.$$

Interpretation. With no information acquisition, the seller still wants the flattest possible payoff subject to limited liability.

3.3 Debt with information acquisition: Proposition 4

If the optimal security at price q induces information acquisition, it is still debt.

Interpretation. A non-flat payoff gives the buyer incentives to distinguish states. Flattening high-state payoffs reduces adverse selection.

3.4 Variable aggregate cash flow: Proposition 5

In a production variant, debt is no longer optimal. The optimal security satisfies $s^*(\theta) = \theta$ for $\theta \leq D^*$ and strictly increases in θ for $\theta > D^*$, with $s^*(\theta) < \theta$.

Interpretation. When information acquisition can create surplus, the seller may want to encourage information rather than suppress it.

3.5 Bounded marginal information cost: Proposition 6

If marginal information cost is bounded, the optimal security is debt on all states where trade occurs with positive probability.

3.6 Multiple underlying assets: Proposition 7

For N assets, the optimal security is debt backed by the pooled cash flow:

$$s^*(\theta_1, \dots, \theta_N) = \min \left\{ \sum_{n=1}^N \theta_n, D^* \right\}.$$

Interpretation. Pooling prevents the buyer from distinguishing individual-asset states with the same total cash flow.

4 Identification and theoretical design

- The result is theoretical, not empirical.
- Flexible information acquisition is the central innovation: the buyer chooses both how much to learn and where to allocate attention.
- Debt is robust in the baseline because any payoff variation across high states invites information acquisition.
- Fixed aggregate cash flow is essential; in production environments information can create surplus and debt may fail to be optimal.
- The paper does not assume monotone securities; debt emerges endogenously.

5 Figures

5.1 Figures 1–3: Payoff and price of a debt security; buyer’s optimal strategy; buyer’s local intensity of information acquisition

Read: Figures 1–3 show that debt makes high states payoff-equivalent. Information acquisition is concentrated around the default-relevant region and falls to zero when the payoff is flat.

Economic message. Debt minimizes information sensitivity by making upside states irrelevant for the buyer’s payoff.

5.2 Figures 4–6: Payoff and price of a call option; buyer’s optimal strategy; buyer’s local intensity of information acquisition

Read: Figures 4–6 show that a call option is sensitive to upside states, so the buyer focuses information acquisition on those high-state regions.

Economic message. Security design changes where sophisticated buyers look; call-like payoffs invite information production about upside states.

5.3 Figures 7–8: Comparative statics with respect to asset opaqueness in booms and busts

Read: In booms, the buyer does not acquire information and higher opaqueness raises face value and price. In busts, the buyer acquires information; higher opaqueness reduces information acquired but lowers face value and price.

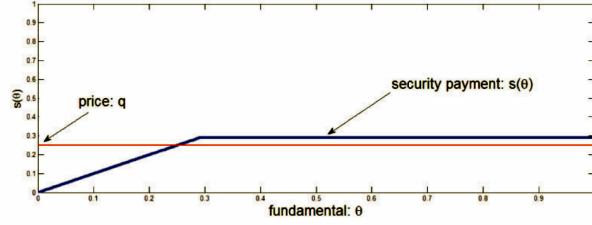


FIGURE 1
Payoff and price of a debt security

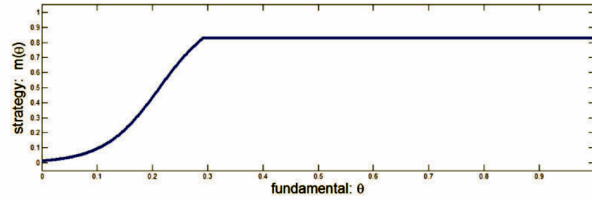


FIGURE 2
Buyer's optimal strategy

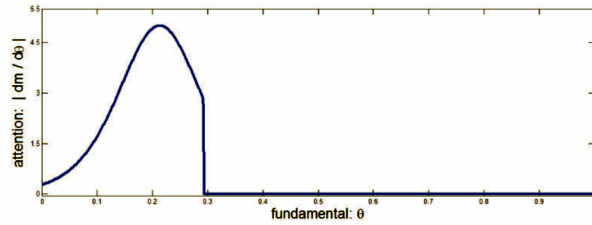


FIGURE 3

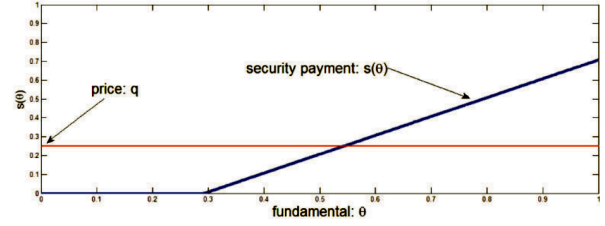


FIGURE 4
Payoff and price of a call option

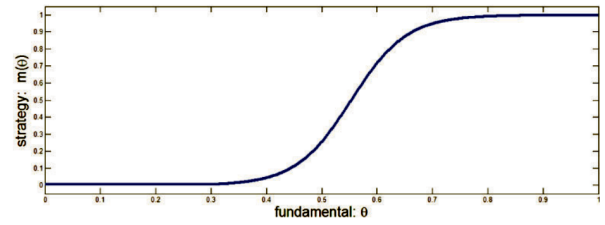
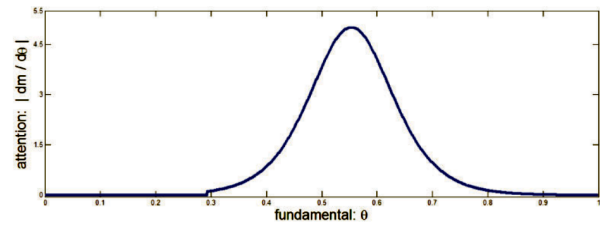


FIGURE 5
Buyer's optimal strategy



Economic message. Opacity supports liquidity in booms but can restrict financing in busts.

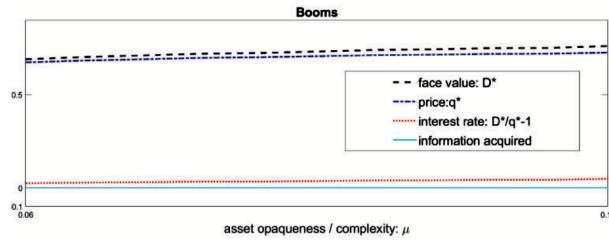


FIGURE 7
Comparative statics w.r.t. asset opacity in booms

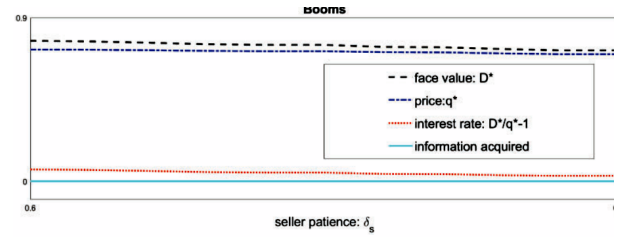
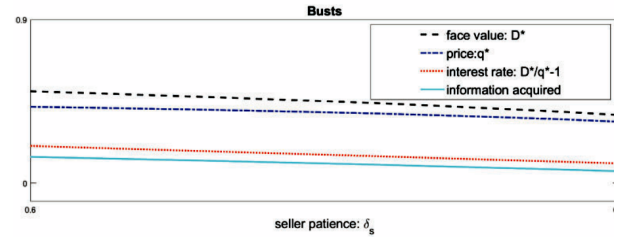
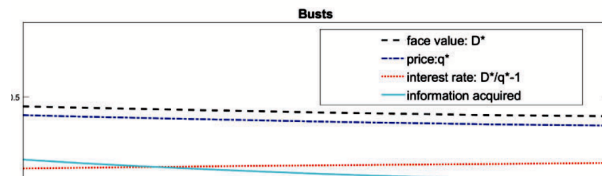


FIGURE 9
Comparative statics w.r.t. seller's patience in booms



5.4 Figures 9–10: Comparative statics with respect to seller patience in booms and busts

Read: Greater seller patience reduces liquidity urgency, so face value and price decline. In busts, information acquisition also declines as gains from trade shrink.

Economic message. Debt terms reflect liquidity urgency: impatient sellers issue larger claims at higher effective yields.

6 Short summary

- The paper studies optimal security design when buyers can flexibly acquire costly information.
- Debt is uniquely optimal in the baseline model because it minimizes information sensitivity.
- The buyer's information acquisition against debt is concentrated around default and zero in high states.
- Call-like securities attract information acquisition about high states.
- In booms, opacity can raise trade scale; in busts, opacity lowers price and face value.
- With multiple assets, the optimal design pools the assets and issues debt backed by the total cash flow.
- Debt may not be optimal when information acquisition creates aggregate surplus.

7 Conclusion

7.1 Core conclusion

Yang shows that debt is uniquely optimal when the seller needs liquidity and the buyer can flexibly acquire costly information before buying. Debt emerges because its flat tail suppresses the buyer's incentive to learn about high states.

7.2 Mechanism

The mechanism is not traditional risk sharing. It is information control. Debt compresses payoff differences across high states and therefore reduces the private value of costly information acquisition.

7.3 Relation to ABS markets

Senior debt tranches in ABS markets are designed to be insensitive to upside collateral details. This model provides a microfoundation for why such claims are liquid when buyers are sophisticated.

7.4 Boundary of the result

When aggregate cash flow is variable and information can create surplus, optimal securities become more state sensitive and debt need not be optimal.

7.5 Takeaway

Financial contracts shape information production. Debt is a liquidity-enhancing contract because it makes the security less worth investigating.